and determine from that exactly *how* distant the galaxy is. (With some technical caveats.)[fn4]

But redshift is *also* connected to cosmic time. The expansion of the universe makes a lot of things weird in astronomy, and one of them is that we use what is essentially a color, written as a number, to denote speed, distance, and "the age the universe was at the time when the thing was shining." Physics is wild.

Here's how it works. If we measure a galaxy's redshift, we know how quickly it's receding from us, and we can use the Hubble-Lemaître Law to get its distance. But because light takes time to travel to us, and we know light's speed, knowing the distance also tells us how long the light has been en route. That means that measuring a galaxy's redshift tells us how long ago the light left the galaxy. And since we know how old the universe is now, that tells us how old the universe was when that galaxy sent out the light we see.

Taking this all into account, astronomers can use redshift to refer to earlier epochs of the universe. "High redshift" is long ago when the universe was young; "low redshift" is more recent. Redshift 0 is the local, present-day universe; redshift 1 is seven billion years ago. At the high end, redshift 6 is a universe only a billion or so years into its life, and the very beginning of the universe, if we could see it, would have a redshift of infinity.

So: a high-redshift galaxy is a distant galaxy that existed when the universe was young, and a low-redshift galaxy is a relatively nearby object living in what is basically the "modern" universe.

The distance-age-redshift relationship is incredibly useful in cosmology. But it relies on the fact that the recession speed always increases with distance in a known way. What if the expansion were to suddenly slow down? What if it were to *stop*, and reverse course? One consequence is that it would completely throw off our distance-measurement rules of thumb and upset a lot of astronomers. Another, nearly as important consequence, depending on who you ask: it would spell doom for the universe and everything in it.

WHAT GOES UP